Module 4.3: Statistical Reasoning - Key Terms & Definitions, PART 3

Statistical Reasoning – Key Terms & Definitions, Part 3

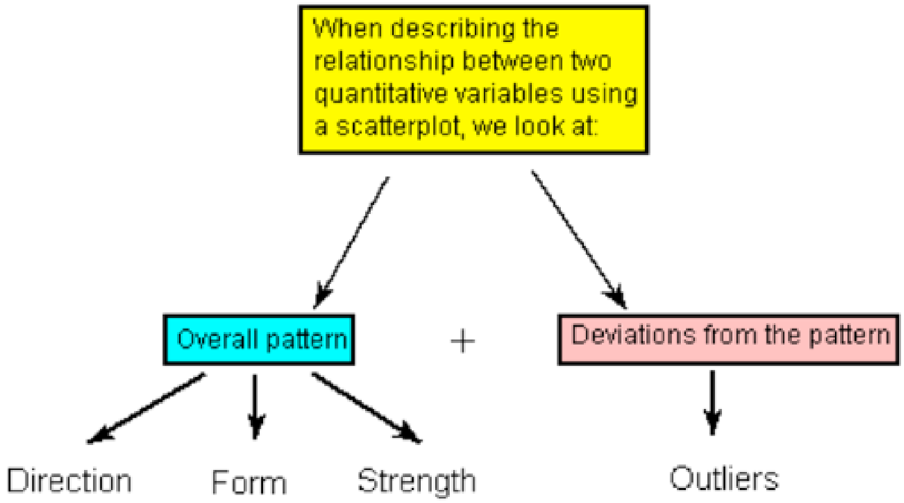
Part 3.1: Case Q -> Q

*We explored the relationship between two variables by comparing the distribution of the **response variable** for each category of the **explanatory variable**:*

- *In case C -> Q, we compared distributions of the quantitative response*
- *In case C -> C, we compared distributions of the categorical response*

***Case Q -> Q** is different in the sense that both variables (in particular the explanatory variable) are quantitative, and, therefore, this case will require a different kind of treatment and tools (-> Display: **scatterplot**).*

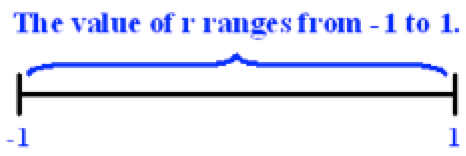
- When describing the relationship as displayed by the ***scatterplot***, be sure to consider:
  - **Overall pattern** -> direction, form, strength
  - Deviations from the pattern -> **outliers**.

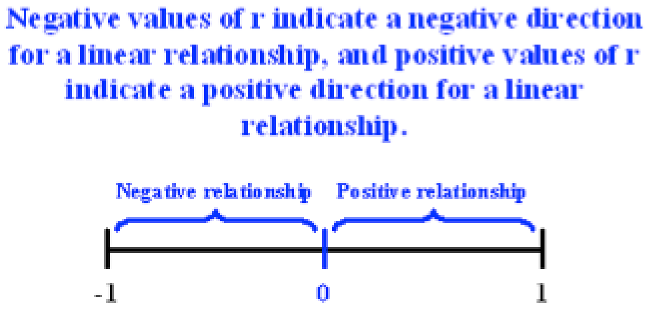


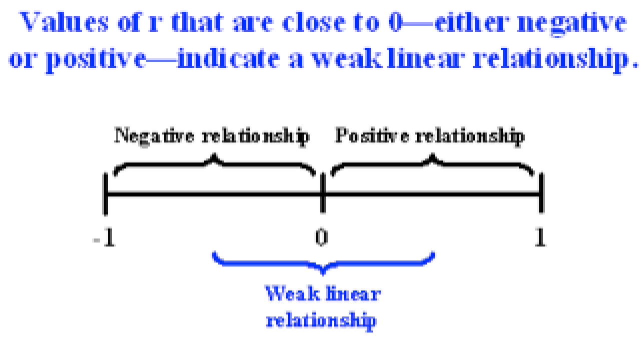
The **Correlation Coefficient – r:** is a numerical measure that measures the **strength** and **direction** of a ***linear*** relationship.

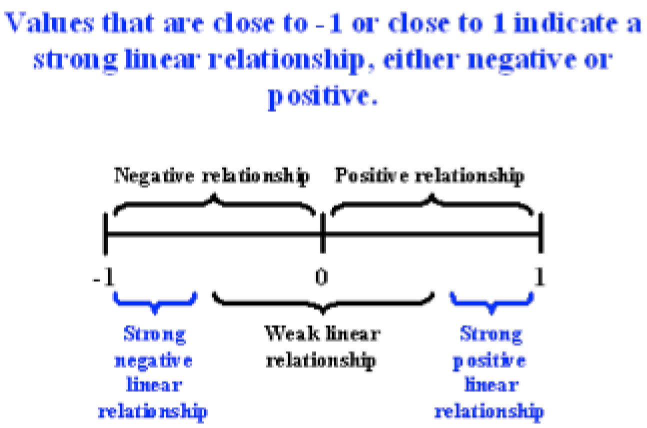
- ***Interpretation:***
- The value of the **Correlation Coefficient – r** ranges from **– 1 to 1**
  - Close to **–1/+1** – **strong**
  - Close to **0** – **weak**
  - Close to **–0.5/+0.5** – **moderately strong** (see below)
    - *Values near – 1 indicate a strong negative linear relationship, values near 0 indicate a weak linear relationship, and values near 1 indicate a strong positive linear relationship.*

  - **Important:** The **Correlation Coefficient – r** can ***only*** be interpreted as the ***measure of the strength of a linear relationship!***



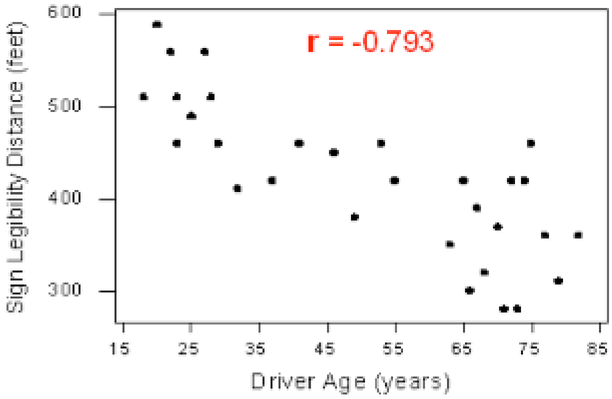




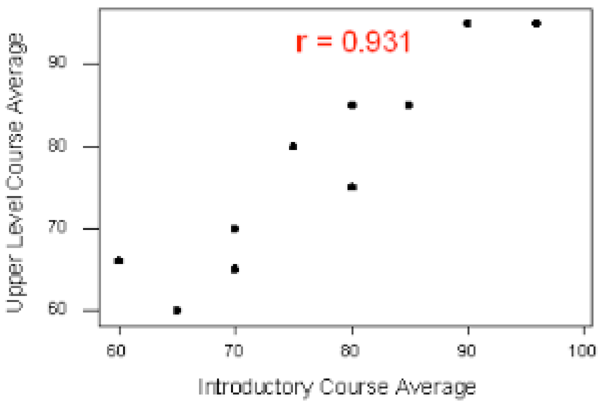


- **Example:** The relationship between the age of the driver and the maximum distance at which a highway sign is legible.

- **The** **scatterplot** suggests a relationship that is **negative** in **direction**, **linear** in **form**, and **moderately strong**.



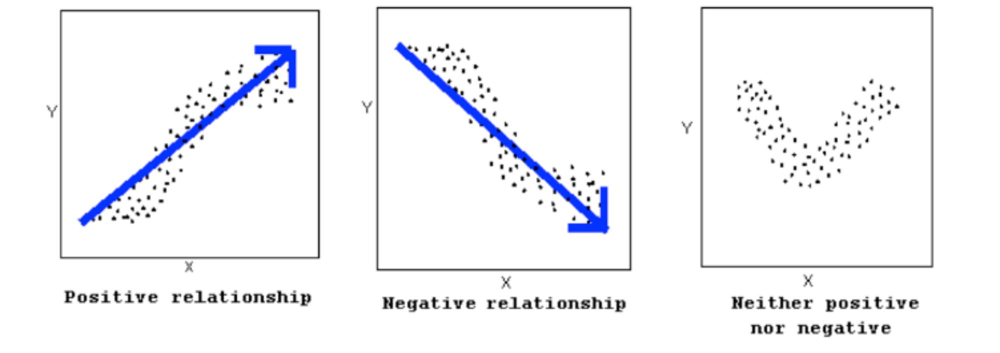
- **The** **scatterplot** suggests a relationship that is **positive** in **direction**, **linear** in **form**, and seems quite **strong**.



- **Important:** **Please take a look at 5.5.15 (pp. 16-18):** The properties of the **Correlation Coefficient – r**
    - *The correlation is only an appropriate numerical measure for linear relationships, and is sensitive to outliers. Therefore, the correlation should only be used as a supplement to a scatterplot (__**after we look at the data**__).*

The **Direction of the Relationship:** The direction of the relationship indicates the relationship's **strength** and **form**.

- The **direction of the relationship** can be **positive**, **negative**, or **neither**:

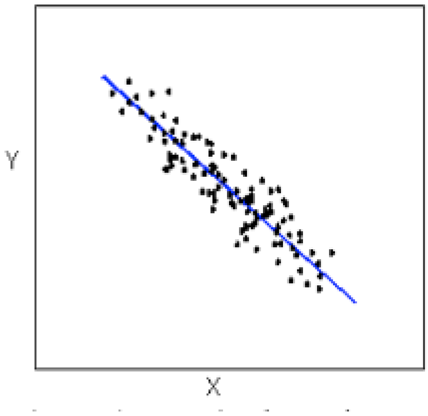


- A **positive (or increasing) relationship** means that an increase in one of the variables is associated with an increase in the other.
- A **negative (or decreasing) relationship** means that an increase in one of the variables is associated with a decrease in the other.
- Not all relationships can be classified as either positive of negative.

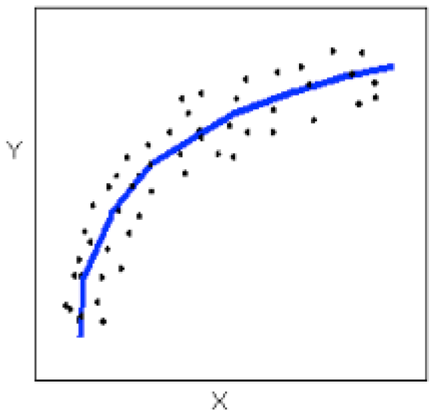
Extrapolation: is prediction of values of the **explanatory variable** that fall outside the range of the data. Since there is no way of knowing whether a relationship holds beyond the range of the explanatory variable in the data, **extrapolation** ***is not reliable and should be avoided.***

The **Form of the Relationship:** The form of the relationship is its general **shape – linear, curvilinear,** and **cluster**.

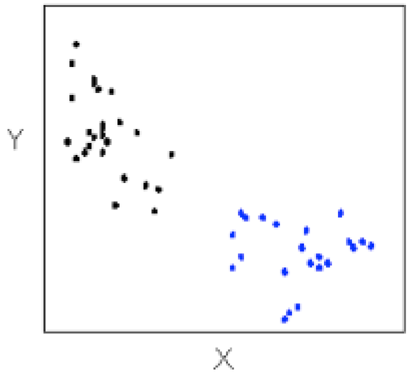
- Relationships with a ***linear form*** are most simply described as points scattered about a line:



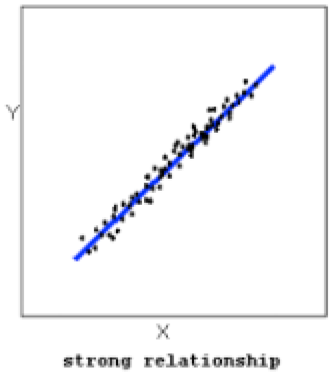
- Relationships with a ***curvilinear form*** are most simply described as points dispersed around the same curved line:

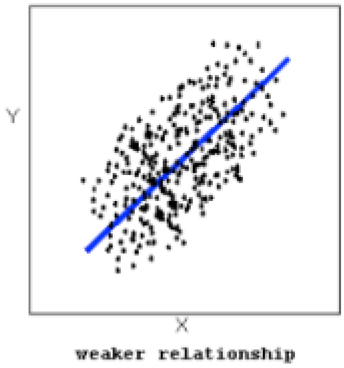


- Another ***form-related pattern*** is ***clusters*** in the data:



- The **strength of the relationship** is determined by how closely the data follow the **form of the relationship**:





Explanatory Variable (Independent Variable): The variable that claims to explain, predict, or affect the response.

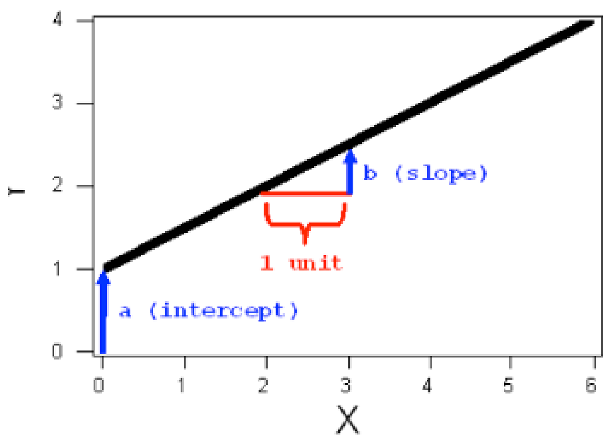
- **Example:** We want to explore whether the outcome of the study – the score on a test – is affected by the test-taker's gender:
  - Explanatory Variable: Gender
  - Response Variable: Test score

Linear Regression: is the technique of finding the line that ***best fits the pattern*** of the ***linear relationship*** **–**the line that best describes how the **response variable** linearly depends on the **explanatory variable**.

- The most commonly used criterion to find the line that ***best fits the pattern*** is called **least squares regression line:** It has the smallest sum of squared vertical deviations of the data points from the line.
- The **least squares regression line** predicts the value of the response variable for a given value of the explanatory variable.
  - The **least-squares regression** for summarizing the ***linear*** relationship between the **response variable (Y)** and the **explanatory variable (X)** has the form**:**

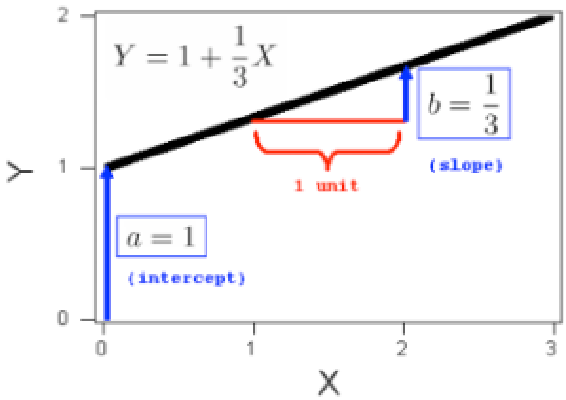
$$Y = a + bX$$

- In this equation, **a** and **b** are constants that can be either negative or positive
- The reason to write the line in this form is that the constants **a** and **b** tell you what the ***line looks like***:
  - The **intercept (a)** is the value that Y takes when X = 0
  - The **slope (b)** is the change in Y for every increase of 1 unit in X.



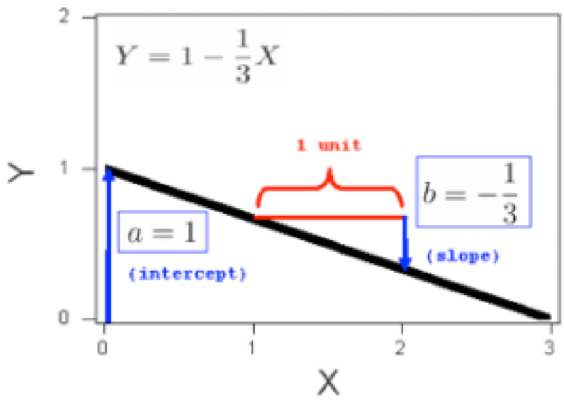
- **Examples:**

Consider the line $Y = 1 + \frac{1}{3}X$. The intercept is 1. The slope is 1/3, and the graph of this l



Example (I)

Consider the line $Y = 1 - \frac{1}{3}X$. The intercept is 1. The slope is −1/3, and the graph of this line is, therefore:



- <u>All you need to do is calculate the **intercept a**, and the **slope b**:</u>

- $\overline{X}$ – the mean of the explanatory variable's values
- S_X – the standard deviation of the explanatory variable's values
- $\overline{Y}$ – the mean of the response variable's values
- S_Y – the standard deviation of the response variable's values
- r – the correlation coefficient

- Given the five quantities above, the **slope** and **intercept** of **the least squares regression line** are found using the following formula:

- $b = r\left(\frac{S_Y}{S_X}\right)$
- $a = \overline{Y} - b\overline{X}$

- **Example:**

Example (J) Age-Distance

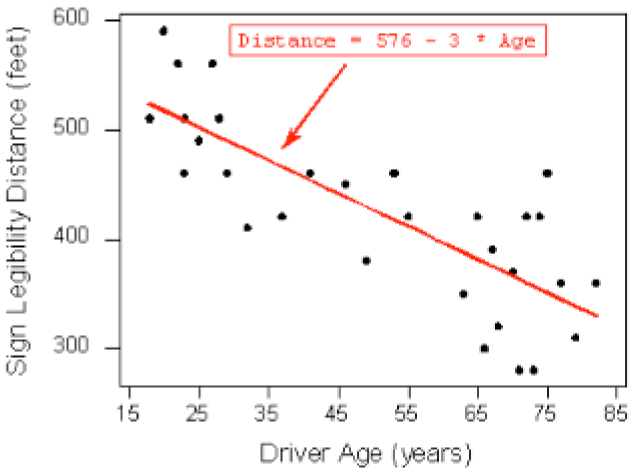
Let's revisit our age-distance example, and find the *least-squares regression line*. The following output will be helpful in getting the 5 values we need:

Statistic	Age (X)	Distance (Y)
Mean	51	423
StDev	21.78	82.8
Correlation	-0.793	

- The *slope* of the line is: $b = -0.793\frac{82.8}{21.78} = -3$. This means that for every 1-unit increase of the explanatory variable, there is, on average, a 3-unit decrease in the response variable. The interpretation *in context* of the slope being –3 is, therefore: For every year a driver gets older, the maximum distance at which he/she can read a sign decreases, *on average*, by 3 feet.
- The *intercept* of the line is: a = 423 – (–3 × 51) = 576 and therefore the *least squares regression line* for this example is:

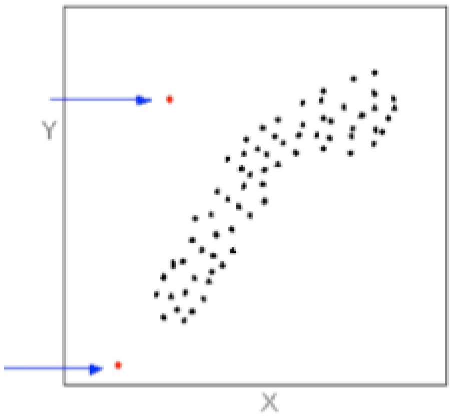
$$\text{Distance} = 576 + (-3 \times \text{Age})$$

Here is the regression line plotted on the scatterplot:



As we can see, the regression line fits the linear pattern of the data quite well.

Outliers: Data points that deviate from the pattern of the relationship:



Overall Pattern: When describing the overall pattern of the relationship between two **quantitative variables (Q -> Q)** we look at its **direction**, **form**, and **strength**. (see above)

Response Variable (Dependent Variable): The outcome of the study.

- **Example:** We want to explore whether the outcome of the study – the score on a test – is affected by the test-takers' gender:
  - Explanatory Variable: Gender
  - Response Variable: Test score

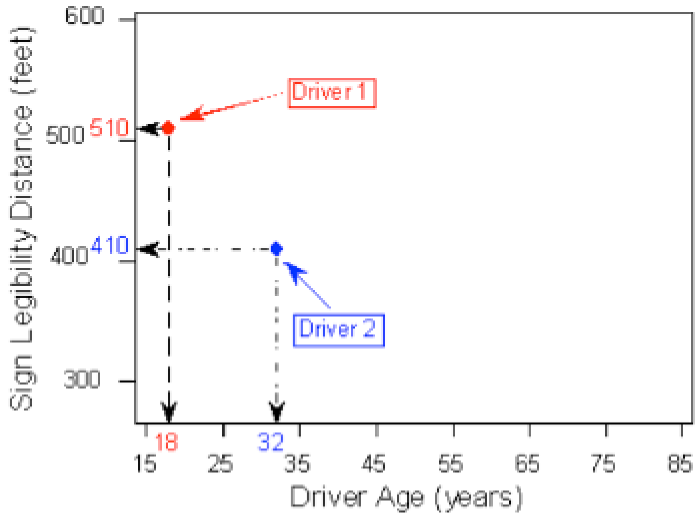
Scatterplot: The graphical display for examining the relationship between two **quantitative variables**.

- Each pair of values is plotted, so that the value of the **explanatory variable (X)** is plotted on the **horizontal axis**, and the value of the **response variable (Y)** is plotted on the **vertical axis**. *(If in a specific example, we do not have a clear distinction between explanatory and response variables, each of the variables can be plotted on either axis*).
  - **Example:** You want to explore the effect of age on maximum legibility

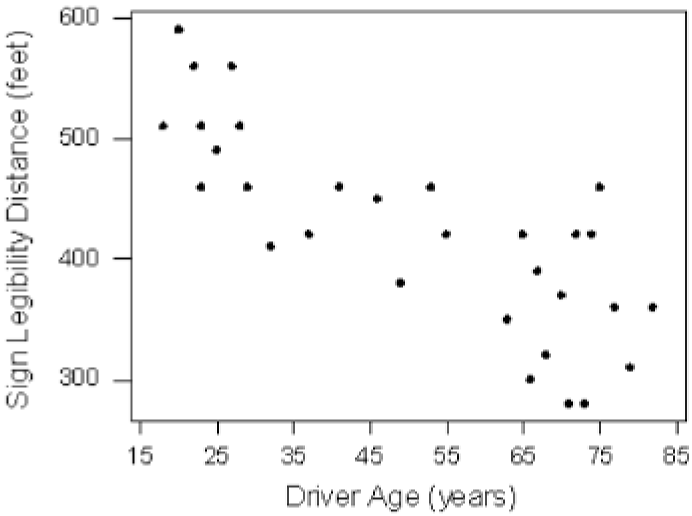
distance:

- the **explanatory variable** is Age
- the **response variable** is Distance
    - Each individual (here, driver) appears on the **scatterplot** as a *single point* whose X-coordinate is the value of the **explanatory variable** for that individual, and whose Y-coordinate is the value of the **response variable** (see below).

	Age (X)	Distance (Y)
Driver 1	18	510
Driver 2	32	410
Driver 3	55	420
Driver 4	23	510
.	.	.
.	.	.
.	.	.
Driver 30	82	360



And here is the completed scatterplot:

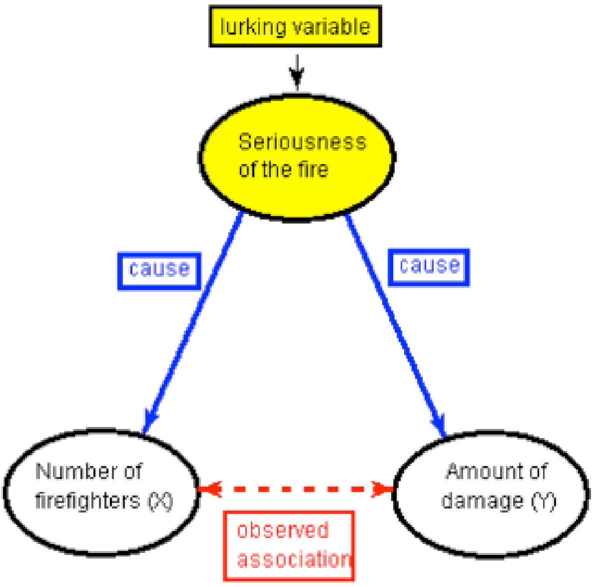


Part 3.2: Causation

- When we explore the relationship between two variables, there is often a temptation to conclude from the observed relationship that changes in the explanatory variable *causes* changes in the response variable. However, **Association does not imply causation, due to the possible presence of lurking variables!**

Lurking Variable (also called a Confounding Variable): A variable that is not among the **explanatory** or **response variables** in a study, but could substantially affect the interpretation of the relationship among those variables.

- See example ***Fire Damage*** (5.6.2, p. 26) in your reading:



- Because of the possibility of **lurking variables**, we adhere to the principle that ***association does not imply causation***
- Including a **lurking variable** in our exploration may
  - Help us gain a deeper understanding of the relationship between variables (see example *Hospital Death Rates*, p. 29)
  - Lead us to rethink the direction of an association (see example *College*

Entrance Exam, pp. 31-33).

- Whenever ***including a lurking variable causes us to rethink the direction of an association***, this is an instance of the ***Simpson's paradox.***